## FLOW DIAGRAM SYMBOLS

- BLIND PADDLE (CLOSED)
- BLIND - SPACER
- BLIND - SPECTACLE NORMALLY CLOSED
- BLIND - SPECTACLE NORMALLY OPEN
- COMPRESSOR CYLINDER
- CONT DRAINER
- DIAPHRAM SEAL
- DRAIN HUB - CLOSED DRAIN SYSTEM
- DRAIN HUB - OPEN DRAIN SYSTEM
- EXPANSION JOINT
- FILL SPOUT
- FREE DRAINING — FREE DRAINING
- FIRE MONITOR
- FIREWATER HYDRANT
- FLAME ARRESTOR
- F — TO FLARE HEADER
- FLEXIBLE HOSE
- FLEXIBLE JOINT
- FLOW ELEMENT (CORIOLIS METER)
- FLOW ELEMENT (THERMAL DISPERSION)
- FLOW ELEMENT (MAGNETIC)
- FLOW ELEMENT (NOZZLE)
- FLOW ELEMENT (ORIFICE FLANGES)
- FLOW ELEMENT PD METER
- FLOW ELEMENT PITOT METER
- FLOW ELEMENT (QUICK CHANGE ORIFICE PLATE)
- FI — FLOW ELEMENT (ROTOMETER)
- FLOW ELEMENT TARGET METER
- FLOW ELEMENT ULTRASONIC METER
- FLOW ELEMENT TURBINE METER
- FLOW ELEMENT (VENTURI)
- FLOW ELEMENT (VORTEX METER)
- HORN
- HOSE REEL
- INSULATING FLANGES
- MEOH — METHANOL
- PIPING - CAP (THREADED OR SOCKET WELD)
- PIPING - CAP (WELDED)
- TP XX — PIPING - CUSTOMER TIE-IN
- PIPING - FLANGED CONNECTION WITH BLIND FLANGE
- PIPING - HOSE CONNECTION
- PIPING - LINE NUMBER BREAK
- XX/XX — PIPING - LINE SPECIFICATION CHANGE
- PIPING - PLUG (THREADED OR SOCKET WELD)
- SP XX — PIPING - SPECIALTY ITEM
- PIPING - UNION
- PSV & PRESSURE - VACUUM MANHOLE
- (P) — PULSATION DAMPENER
- REDUCER - LINE SIZE CHANGE
- RUPTURE DISK
- RUPTURE DISK (SAFETY HEAD)
- SIGHT/FLOW GLASS
- SILENCER
- SLOPE
- SPRAY NOZZLE
- T — STEAM TRAP
- T — STEAM TRAP/STRAINER COMBINATION
- STRAIGHTENING VANES
- STRAINER (BUCKET TYPE)
- STRAINER (CONE TYPE)
- STRAINER ('T' TYPE)
- 3/4" — STRAINER ('Y' TYPE)
- VENT WITH INSECT SCREEN
- VENT
- VENDOR BOX

## OFF PAGE CONNECTORS

- XXX / BL / TO/FROM — OFF UNIT CONNECTION SYMBOL
- XXX / XXX / TO/FROM — OFF DRAWING CONNECTION SYMBOL
- XXX / XXX / TO/FROM — OFF DRAWING BI-DIRECTION CONNECTION SYMBOL

## FLOW DIAGRAM LINES

- PROCESS LINE ARROW
- EXISTING OR FUTURE LINES
- PRIMARY PROCESS LINES
- UTILITY AND SECONDARY PROCESS LINES
- STEAM — JACKETED LINES
- ELECTRIC/STEAM TRACED LINES
- TUBING

## VALVES

- ANGLE VALVE
- BALL VALVE
- V-BALL VALVE
- INTERMITTENT BLOWDOWN VALVE
- CONTINUOUS BLOWDOWN VALVE
- BUTTERFLY VALVE
- CSO/CSC — CAR SEAL OPEN/ CAR SEAL CLOSED
- CHECK VALVE
- CHOKE VALVE
- CHOKE VALVE-ADJUSTABLE
- DUMP VALVE (SPRING LOADED)
- EB — EXTENDED BONNET VALVE
- 3-WAY VALVE
- 4-WAY VALVE
- FRICK GVD VALVE
- GATE VALVE
- GLOBE VALVE
- HAND CONTROL VALVE
- IN-LINE PRESSURE SAFETY VALVE
- NEEDLE VALVE
- (M) — MOTOR OPERATED VALVE
- VALVE - NORMALLY CLOSED
- VALVE - NORMALLY OPEN
- PISTON OPERATED VALVE
- PLUG VALVE
- PNEUMATIC CONTROL VALVE
- PNEUMATIC CONTROL VALVE WITH HAND WHEEL
- PNEUMATIC CONTROL VALVE WITH ADJUSTABLE MECHANICAL MAXIMUM STOP
- PNEUMATIC CONTROL VALVE WITH ADJUSTABLE MECHANICAL MINIMUM STOP
- PNEUMATIC CONTROL VALVE WITH ADJUSTABLE MECHANICAL MINIMUM AND MAXIMUM STOP
- PRESSURE RELIEVING REGULATOR (SELF CONTAINED)
- PRESSURE REDUCING REGULATOR (SELF CONTAINED)
- PSV PRESSURE - VACUUM
- PSV VACUUM
- PRESSURE SAFETY VALVE (PILOT OPERATED)
- M — PRESSURE SAFETY VALVE (M-MODULATING, S-SNAP ACTING)
- PRESSURE SAFETY VALVE (SPRING OPERATED)
- SEAL CAP ACTUATOR (MOUNTED AT TOP SIDE OR BOTTOM OF ACTUATED DEVICE AS APPLICABLE)
- S — SOLENOID OPERATED VALVE
- S — SOLENOID VALVE (3-WAY)
- S — SOLENOID VALVE (4-WAY)
- S / R — SOLENOID VALVE WITH MANUAL RESET
- INDICATES PATH IN FAILED POSITION
- STOP CHECK VALVE
- TEST OR BLEED RING
- V-BALL CONTROL VALVE

## LINE CLASS DESIGNATION

| LINE CLASS | BASE MATERIAL | ANSI CLASS | FLANGE TYPE | CA | PRESSURE/TEMP RANGE | SERVICE |
|---|---|---|---|---|---|---|
| 1CB1S01 | CARBON STEEL (KILLED) | 150 | RF | 0.063 | 285 PSIG @ -20 TO 100°F THROUGH 170 PSIG @ 500°F | GENERAL PURPOSE PROCESS, STEAM, CONDENSATE, NITROGEN, COOLING WATER |
| 3CB1S01 | CARBON STEEL (KILLED) | 300 | RF | 0.063 | 740 PSIG @ -20 TO 100°F THROUGH 410 PSIG @ 800°F | GENERAL PURPOSE PROCESS, STEAM, CONDENSATE |
| 6CB1S01 | CARBON STEEL (KILLED) | 600 | RF | 0.125 | 1480 PSIG @ -20 TO 100°F THROUGH 825 PSIG @ 800°F | HIGH PRESSURE, LOW TEMPERATURE SYNGAS |
| 6CJ1S01 | 1-1/4 Cr- 1/2 Mb | 600 | RF | 0.120 | 1500 PSIG @ -20 TO 100°F THROUGH 1015 PSIG @ 800°F | HIGH TEMPERATURE PROCESS, HYDROGEN SERVICE |
| 9CJ1S01 | 1-1/4 Cr- 1/2 Mb | 900 | RTJ | 0.120 | 2250 PSIG @ -20 TO 100°F THROUGH 1525 PSIG @ 800°F | HIGH PRESSURE, HIGH TEMPERATURE SYNGAS |
| 1SA0S01 | 304/304L SS | 150 | RF | 0.031 | 275 PSIG @ -20 TO 100°F THROUGH 170 PSIG @ 500°F | GENERAL PROCESS REQUIRING CORROSION RESISTANCE |
| 3SA0S01 | 304/304L SS | 300 | RF | 0.031 | 720 PSIG @ -20 TO 100°F THROUGH 355 PSIG @ 1000°F | GENERAL PROCESS REQUIRING CORROSION RESISTANCE |
| 33SC0S01 | 304H SS | 300 | RF | 0.031 | 720 PSIG @ -20 TO 100°F THROUGH 40 PSIG @ 1500°F | HIGH TEMPERATURE PROCESS REQUIRING CORROSION RESISTANCE |
| 6SA0S01 | 304/304L SS | 600 | RF | 0.031 | 1440 PSIG @ -20 TO 100°F THROUGH 710 PSIG @ 1000°F | GENERAL PROCESS REQUIRING CORROSION AND/OR HEAT RESISTANCE |
| 12CS1T01 | CARBON STEEL | 125 | RF | 0.063 | 150 PSIG @ -20 TO 150°F | UTILITY (AIR AND WATER), CATEGORY D, GALV. |
| 1CS1S02 | CARBON STEEL | 150 | RF | 0.063 | 285 PSIG @ -20 TO 800°F | ELEVATED TEMPERATURE FLUID SERVICE, STEAM/CONDENSATE |
| 3CS1S02 | CARBON STEEL | 300 | RF | 0.063 | 740 PSIG @ -20 TO 800°F | ELEVATED TEMPERATURE FLUID SERVICE, STEAM/CONDENSATE |

## LINE IDENTIFICATION

XX" - XXX - XXXXXXX - XXXX - XX

- LINE SIZE
- SERVICE
- LINE CLASS (SEE LINE CLASS TABLE ABOVE)
- LINE NUMBER
- INSULATION TYPE
  - ET = ELECTRIC TRACED
  - HC = HEAT CONSERVATION
  - NI = NOT INSULATED
  - PP = PERSONNEL PROTECTION
  - ST = STEAM TRACED

## LINE SERVICE IDENTIFICATION

### PROCESS SERVICE

| | | | | | |
|---|---|---|---|---|---|
| BIOGAS | BG | HYDROGEN | H | NATURAL GAS | NG |
| BUTANE (FULL RANGE) | BUT | ISO-BUTANE | IBU | NATURAL GAS LIQUIDS | NGL |
| CRUDE | CRD | JET FUEL | JET | N-BUTANE | NBU |
| DIESEL | DSL | KEROSENE | KER | OFFGAS | OFG |
| GAS OIL (ATMOSPHERIC) | AGO | LIGHT STRAIGHT RUN | LSR | PROCESS CONDENSATE | PC |
| GAS OIL (LIGHT) | LGO | MIXED GAS | MFG | PROPANE | PRO |
| GAS OIL (HEAVY) | HGO | NAPHTHA (FULL RANGE) | NAP | REFINERY FUEL GAS | RFG |
| GAS OIL (VACUUM) | VGO | NAPHTHA (HEAVY) | NPH | RESID | RSD |
| GENERAL PROCESS | P | NAPHTHA (LIGHT) | NPL | SYNGAS | SG |
| HYDROCARBON | HC | | | | |

### MISCELLANEOUS

| | | | | | |
|---|---|---|---|---|---|
| BLOWDOWN | BD | SLOP | SLO | WASTE WATER | WW |
| EQUIPMENT TRIM | VT | VENT | V | | |

### UTILITIES

| | | | | | |
|---|---|---|---|---|---|
| AIR - INSTRUMENT | IA | DRAIN (CLOSED) | DRC | POTABLE WATER | POW |
| AIR - PLANT | PAS | DRAIN (OPEN) | DRO | STEAM (HP) | HS |
| AIR - UTILITY | UAS | FLARE | FL | STEAM (LP) | LS |
| AMMONIA | NH3 | FUEL GAS | FG | STEAM (MP) | MS |
| BOILER FEED WATER | BFW | HEATING OIL RETURN | HOR | STEAM CONDENSATE | SC |
| CHEMICAL | CHE | HEATING OIL SUPPLY | HOS | STEAM CONDENSATE (HP) | CHP |
| COOLING WATER RETURN | CWR | LUBE OIL | LO | STEAM CONDENSATE (LP) | CLP |
| COOLING WATER SUPPLY | CWS | NITROGEN | N | STEAM CONDENSATE (MP) | CMP |
| DEMINERALIZED WATER | DMW | OILY WATER DRAIN (CLOSED) | OWD | TEMPERED WATER | TW |
| DESALTER WATER | DWA | OILY WATER SEWER (OPEN) | OWS | UTILITY WATER | UTW |

### SOLVENTS

| | |
|---|---|
| AMINE (LEAN) | AML |
| AMINE (RICH) | AMR |
| CAUSTIC (RICH) | CAR |
| CAUSTIC (SPENT) | CAS |
| GLYCOL | GL |
| GLYCOL / WATER RETURN | GWR |
| GLYCOL / WATER SUPPLY | GWS |

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DISTILLATION PIPING AND INSTRUMENTATION DIAGRAM

| | |
|---|---|
| PROJECT: | NEW FRANKEN 1 |
| DRAWING #: | DISTILLATION |
| SHEET #: | 001 |
| DATE: | 06-23-2025 |

## INSTRUMENT FUNCTION LETTERS

(USED INSIDE INSTRUMENT BUBBLES)

SELECT FIRST LETTER FROM TABLE AT LEFT

| | | | |
|---|---|---|---|
| A | ANALYSIS | • A | ALARM |
| B | BURNER/COMBUSTION | • AL | ALARM LOW |
| C | CONDUCTIVITY (ELECT.) | • ALL | ALARM LOW (SHUTDOWN) |
| E | VOLTAGE (EMF) | • AH | ALARM HIGH |
| F | FLOW RATE | • AHH | ALARM HIGH (SHUTDOWN) |
| FQ | FLOW QUANTITY | • C | CONTROLLER |
| FF | FLOW RATIO | • CC | CYCLE CONTROLLER |
| G | GAUGING (DIMENSIONAL) | • CV | CONTROL VALVE (SELF ACTUATED) |
| H | HAND (MANUALLY) | • CCV | CYCLE CONTROL VALVE |
| I | CURRENT | • E | ELEMENT (PRIMARY) |
| J | SCAN (1) OR POWER (2) | • G | GAUGE OR GLASS |
| K | TIME | • I | INDICATOR |
| L | LEVEL | • IC | INDICATING CONTROLLER |
| M | MOISTURE/HUMIDITY | • IS | INDICATING SWITCH |
| N | SHUTDOWN | • IT | INDICATING TRANSMITTER |
| O | UNCLASSIFIED (AS REQUIRED) | • JI | MULTI-POINT INDICATOR |
| P | PRESSURE/VACUUM | • JR | MULTI-POINT RECORDER |
| PD | PRESSURE DIFFERENTIAL | • K | AUXI. CONTROL STATION (EX.: AUTO/MANUAL) |
| Q | QUALITY (INTEGRATE OR TOTALIZE) | • L | LIGHT |
| R | RADIOACTIVITY | • O | ORIFICE RESTRICTION |
| S | SPEED/FREQUENCY (1) OR SEAL (2) | • P | POINT (TEST CONNECTION) |
| T | TEMPERATURE | • QI | INTEGRATE OR TOTALIZE INDICATOR |
| TD | TEMPERATURE DIFFERENTIAL | • QR | INTEGRATE OR TOTALIZE RECORDER |
| U | MULTIVARIABLE | • R | RECORDER |
| V | VISCOSITY (1) OR VIBRATION (2) | • RC | RECORDING CONTROLLER |
| W | WEIGHT/FORCE | • RT | RECORDING TRANSMITTER |
| WD | WEIGHT DIFFERENTIAL | • S | SWITCH |
| X | UNCLASSIFIED (AS REQUIRED) | • SL | SWITCH LOW (ALARM) |
| Y | COMPUTING FUNCTION | • SLL | SWITCH LOW (SHUTDOWN) |
| Z | EMERGENCY ACTION (1) OR POSITION (2) | • SH | SWITCH HIGH (ALARM) |
| | | • SHH | SWITCH HIGH (SHUTDOWN) |
| | | • SH/L | COMBINED SWITCH HIGH/LOW |
| | | • SE | SAFETY ELEMENT |
| | | • SV | SAFETY RELIEF VALVE |
| | | • T | TRANSMITTER (BLIND) |
| | | • V | VALVE |
| | | • W | WELL OR PROBE |
| | | • Y | RELAY |
| | | • Z | ACTUATOR |
| | | *** | FOR SYMBOLS NOT SHOWN AND FOR FURTHER REFERENCE SEE ISA STD. ISA-S5.1 |

## ABBREVIATIONS (GENERAL)

| | | | |
|---|---|---|---|
| AG | ABOVE GROUND | ISBL | INSIDE BATTERY LIMITS |
| ATM | ATMOSPHERE | LC | LOCK CLOSE |
| BD | BLOWDOWN | LFSD | LOW-FIRE SHUTDOWN |
| BF | BLIND FLANGE | LO | LOCK OPEN |
| B.F. | BACK FLUSH | LP | LOW PRESSURE |
| BL | BATTERY LIMITS | MIN. | MINIMUM DISTANCE |
| BSD | BURNER SHUTDOWN | MW | MANWAY |
| CB | CONTINUOUS BLOWDOWN | MWP | MAXIMUM WORKING PRESSURE |
| CC | CORROSION COUPON | MWT | MAXIMUM WORKING TEMPERATURE |
| CI | CHEMICAL INJECTION | NC | NORMALLY CLOSED |
| CSC | CAR SEAL CLOSE | MAWP | MAXIMUM ALLOWABLE WORKING PRESSURE |
| CSO | CAR SEAL OPEN | NCMH | NORMAL CUBIC METER PER HOUR |
| C&W | COATED AND WRAPPED | NGL | NATURAL GAS LIQUIDS |
| DE | DE-ENERGIZED | NILL | NORMAL INTERFACE LIQUID LEVEL |
| DP | DESIGN PRESSURE | NNF | NORMALLY NO FLOW |
| $\Delta$P | DIFFERENTIAL PRESSURE | NO | NORMALLY OPEN |
| (F) | FURNISHED BY VENDOR | NLL | NORMAL LIQUID LEVEL |
| (E) | EXISTING | OB | OUTLET BASKET |
| E/E | END TO END | OD | OUTSIDE DIAMETER |
| ES | EMERGENCY STOP | OSBL | OUTSIDE BATTERY LIMITS |
| EW | EYE WASH | OWS | OILY WATER SEWER |
| EXT. | EXTERNAL | PB | PUSH BUTTON |
| FC | FAIL CLOSED | PSI | POUNDS PER SQUARE INCH |
| FL | FAIL LAST | PSIG | POUNDS PER SQUARE INCH GAUGE |
| FMU | FITTING MAKE-UP | RTD | RESISTANCE TEMPERATURE DETECTOR |
| FO | FAIL OPEN | RSD | REACTOR SHUTDOWN |
| FOB | FLAT ON BOTTOM | S/FF | SEAM TO FACE OF FLANGE |
| FOT | FLAT ON TOP | SC | SAMPLE CONNECTION |
| FP | FULL PORT VALVE | SEL | SELECTOR SWITCH |
| FV | FULL VACUUM | SO | STEAM OUT |
| GC | GAS CHROMATOGRAPH | SP | SET POINT |
| GO | GEAR OPERATOR | STS | SAFETY SHOWER |
| HC | HOSE CONNECTION | SS | STAINLESS STEEL |
| HH | HAND HOLE | S/S | SEAM TO SEAM |
| HOA | HAND-OFF-AUTO SELECTOR SWITCH | T/T | TANGENT TO TANGENT |
| HP | HORSE POWER | T/FF | TANGENT TO FACE OF FLANGE |
| H.PT. | HIGH POINT | TEG | TRIETHYLENE GLYCOL |
| HS | HAND SWITCH | TSO | TIGHT SHUTOFF |
| HV | HALF VACUUM | UG | UNDERGROUND |
| IA | INSTRUMENT AIR SUPPLY | UL | UNIT LIMITS |
| IB | INTERMITTANT BLOWDOWN | UC | UTILITY CONNECTION |
| I.D. | INSIDE DIAMETER | US | UTILITY STATION |
| IP | INTERMEDIATE PRESSURE | VB | VORTEX BREAKER |

## INSTRUMENT IDENTIFICATION

### INSTRUMENT LINE SYMBOLS

CONNECTION TO PROCESS
MECHANICAL LINK OR INSTR. SUPPLY
INSTRUMENT AIR SIGNAL
ELECTRICAL SIGNAL
DATA HIWAY / SOFT LINK
CAPILLARY TUBING
INFRARED SIGNAL
INSTRUMENT ARROW

### INSTRUMENT SYMBOL DESCRIPTORS

F
X
Y

F = RELAY FUNCTION - X, -, +, <, >, BIAS, ETC.
X = RELAY TAG - LY, TY, PY, FY, ETC.
Y = TAG NUMBER - 1001, 2001, 3001, 4001, ETC.

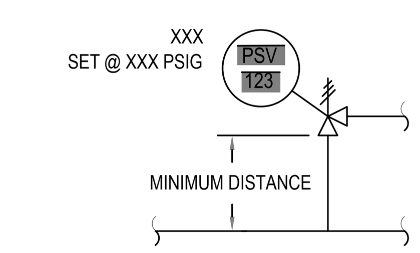
| | |
|---|---|
| AVERAGE FUNCTION | SUMMATION FUNCTION |
| CURRENT TO PNEUMATIC | SQUARE ROOT FUNCTION |
| HOA HAND OFF/AUTO SWITCH | HIGH SELECTOR |
| X MULTIPLICATION FUNCTION | LOW SELECTOR |
| DIFFERENCE FUNCTION | DIVIDING FUNCTION |

HH = PROGRAMMABLE ALARM HIGH HIGH
H = PROGRAMMABLE ALARM HIGH
L = PROGRAMMABLE ALARM LOW
LL = PROGRAMMABLE ALARM LOW LOW

X = SIS, BMS, PLC, RTU, PCS, SCS
Y = TAG NUMBER - 1001, 1002, 1003, ETC.
SIS = SAFETY INSTRUMENTED SYSTEM
SCS = SAFETY CONTROL SYSTEM
BMS = BURNER MANAGEMENT SYSTEM
RTU = REMOTE TERMINATION SYSTEM
PCS = PROCESS CONTROL SYSTEM - LOGIC

## GENERAL NOTES

1. ALL PIPING VENTS ARE 3/4", EXCEPT AS NOTED.
2. ALL PIPING DRAINS ARE 3/4" EXCEPT AS NOTED.
3. CONTROL VALVE ACTION ON AIR OR ELECTRICITY FAILURE: FO-OPENS, FC-CLOSES, FLC- LOCKS STATIC (AND WOULD FINALLY DRIFT TO CLOSE ON LEAKAGE OF AIR FROM ACTUATOR).
4. ALL RV'S AND VENTS DISCHARGING TO ATMOSPHERE SHOULD HAVE 3MM WEEP HOLE AT LOW POINTS. WEEP HOLE TO BE DIRECTED AWAY FROM AREAS WHERE PERSONNEL MAY BE PRESENT.
5. UNLESS OTHERWISE NOTED, LETTER SYMBOLS FOR INSTRUMENTATION CONFORMS TO I.S.A. ENGINEERING STANDARD S5.1
6. DEFINITIONS:
   GRAVITY FLOW - ELEVATIONS DOWNSTREAM NEVER EXCEED INLET ELEVATIONS. LINE MAY CONTAIN POCKETS
   SLOPE - ELEVATION CHANGES ARE DOWNWARD ONLY. NO POCKETS ARE PERMITTED. SPECIFIC SLOPE REQUIRED ARE SHOWN BY SYMBOL WHERE X HAS UNITS THE SAME AS I
   NO POCKETS - NO LIQUID POCKETS IN THE LINE.
7. LOCATION OF RELIEF VALVES SHALL BE:




RELIEF SET POINTS SHALL BE INDICATED FOR ALL RELIEF LINES.
ALL VAPOR SERVICE RELIEF VALVES SHALL BE INSTALLED OFF TOP VAPOR LINES. ALL RELIEF VALVE DISCHARGES SHALL ENTER THE TOP OF VENT/FLARE HEADERS.

8. VALVES IDENTIFIED WITH LO OR LC ARE FOR SAFETY PROTECTION (EG. TO PREVENT BLOCKED-IN CONDITIONS). VALVE OPERATION IS BY AUTHORIZED PROCEDURE.

## LOCAL INSTRUMENT TAG

XX - XXX
- NUMBER IN SEQUENCE (BEGIN EACH FUNCTION WITH NUMBER 001)
- FUNCTION LETTERS

## INSTRUMENT LOOP IDENTIFICATION

XXXX - XXX X XXX X X
- SUFFIX NUMBER (IF REQUIRED)
- SUFFIX LETTER (IF REQUIRED)
- NUMBER IN SEQUENCE (MIN. 2 DIGITS)
- SUFFIX LETTER (IF REQUIRED)
- P&ID NUMBER (MIN. 2 DIGITS)
- ISA FUNCTION LETTERS

## INSTRUMENT SYMBOLS (BUBBLES)

LIGHT
FLOW COMPUTER (BY CLIENT)
LOCAL INTERLOCK
I INTERLOCKING FUNCTION W/IN EMERGENCY SHUTDOWN
P PACKAGE INTERLOCK LOGIC
FIELD MOUNTED INSTRUMENT
LOCAL PANEL MOUNTED INSTRUMENT (FRONT)
LOCAL PANEL MOUNTED INSTRUMENT (REAR)
CONTROL ROOM PANEL MOUNTED INSTRUMENT (FRONT)
CONTROL ROOM PANEL MOUNTED INSTRUMENT (REAR)
ET INSTRUMENT WITH HEAT TRACING
ET = ELECTRICAL TRACING
ST = STEAM TRACING
(DCS) DISTRIBUTED CONTROL SYSTEM (ACCESSIBLE TO OPERATOR)
(DCS) DISTRIBUTED CONTROL SYSTEM (INACCESSIBLE TO OPERATOR)
ESD, BMS, PLC, PCS, SCS AND RTU INTERLOCK (ACCESSIBLE TO OPERATOR)
ESD, BMS, PLC, PCS, SCS AND RTU INTERLOCK (INACCESSIBLE TO OPERATOR)
XX THREADED INSTRUMENT CONNECTION TO PROCESS LINE AND EQUIPMENT
XX SOCKET WELDED INSTRUMENT CONNECTION TO PROCESS LINE AND EQUIPMENT
XX FLANGED INSTRUMENT CONNECTION TO PROCESS LINE AND EQUIPMENT
XX GENERIC INSTRUMENT CONNECTION TO PROCESS LINE AND EQUIPMENT

A
B

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IRON SPONGE TANK PIPING AND INSTRUMENTATION DIAGRAM

| | |
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| PROJECT: | NEW FRANKEN 1 |
| DRAWING #: | IRON SPONGE |
| SHEET #: | 002 |
| DATE: | 06-23-2025 |


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IRON SPONGE TANK
PIPING AND
INSTRUMENTATION
DIAGRAM

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| DRAWING #: | IRON SPONGE |
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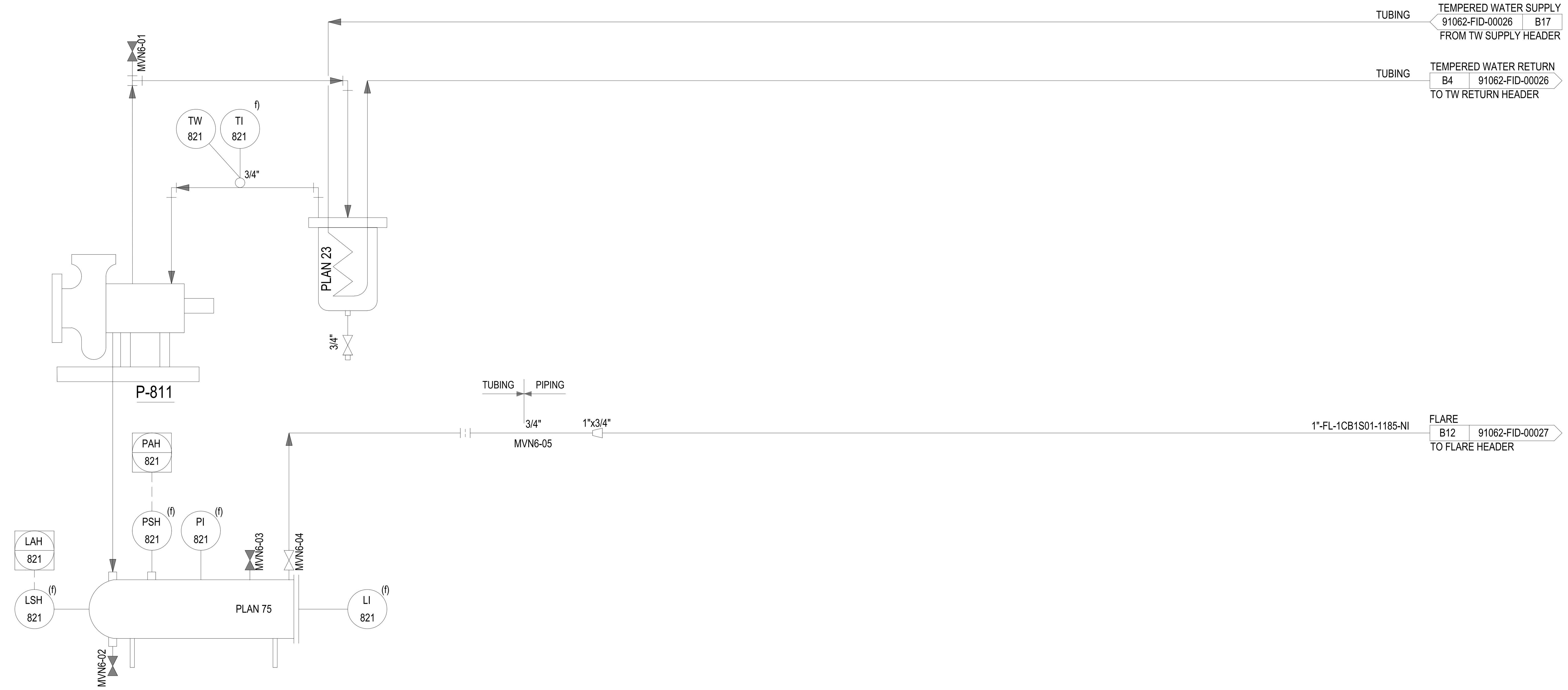
IRON SPONGE TANK
PIPING AND
INSTRUMENTATION
DIAGRAM

| | |
|---|---|
| PROJECT: | NEW FRANKEN 1 |
| DRAWING #: | IRON SPONGE |
| SHEET #: | 004 |
| DATE: | 06-23-2025 |

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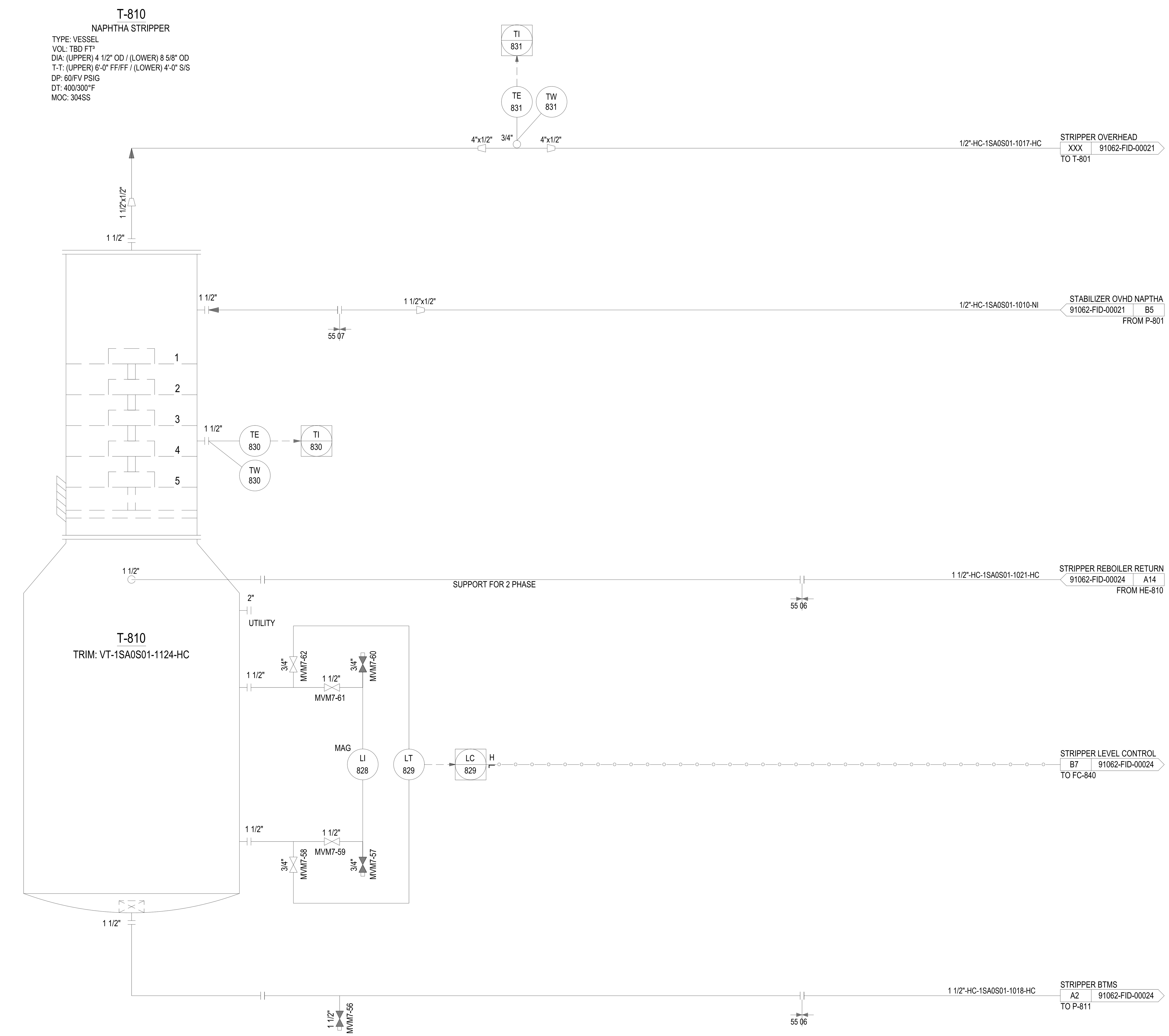


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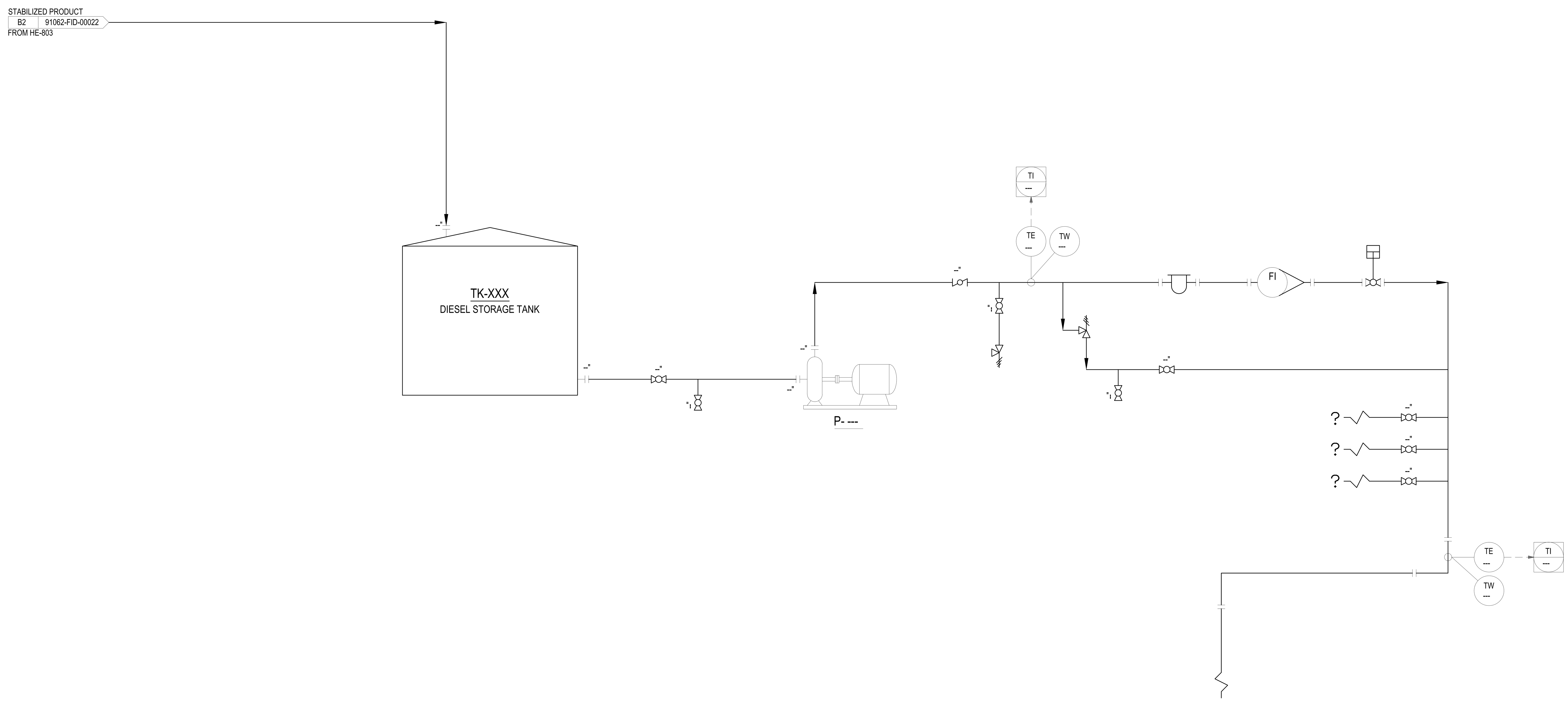


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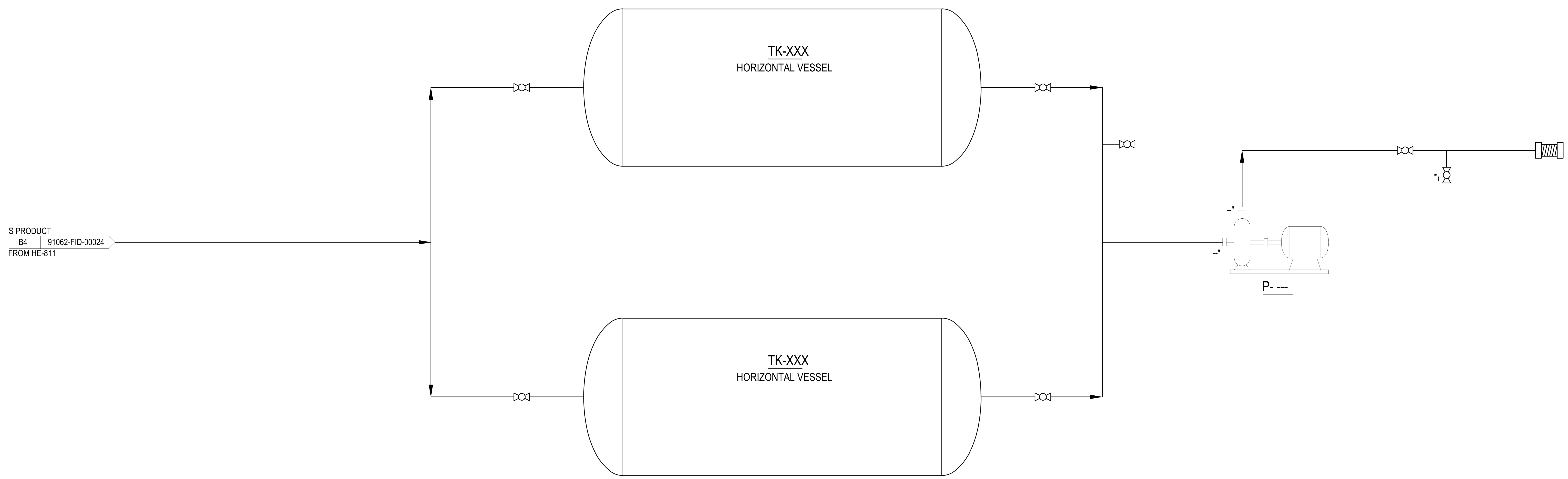


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